DTI PROJECT

Laser Security system

By:V.Sai Sri Nischitha

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CONTENTS

S.No	Title	Page Number
1	Abstract	3
2	Empathy	4
3	Define	5
4	Ideate	
5	Prototype	
6	Blockdiagram& Flowchart	
7	Testing	
8	Conclusion	

ABSTRACT

The Smart Laser Security System is a prototype developed to enhance restricted-area surveillance through cost-effective laser-based intrusion detection. By integrating laser diodes, light-dependent sensors (LDRs), and a microcontroller (e.g., Arduino or ESP8266), the system detects unauthorized entry when a laser beam is interrupted. The design supports both standalone and IoT-enabled configurations, enabling real- time alerts via buzzer and Wi-Fi-based notifications (using platforms like Blynk or IFTTT). This project combines practical circuit design, embedded software logic, and wireless communication to deliver a lightweight, scalable security solution. Rigorous testing confirmed its reliability, responsiveness, and potential for real-world applications in home, lab, and industrial settings.

EMPATHY

Empathy in Design– Laser Security System A smart security system should do more than detect intrusions—it should care about the people using it. That means:

- Easy to use for everyone, even non-technical users.
- Peace of mind with reliable alerts and fewer false alarms.
- Inclusive design for the elderly ,kids ,or people with disabilities.
- Real-life ready by handling petsor everyday movement smartly.

Empathy ensures the system feels like a helpful companion ,not just a machine.

DEFINE

In this step ,you focus on what you're solving and why it matters. For the Smart Laser Security System ,the problem is: “How can we design an affordable ,simple-to-use, and accurate security system that alerts users when someone enters stricted area—without causing panic through false alarms?” You’re defining the need for:

- Accurate intrusion detection Low cost and ease of setup
- Reliable operationin real-world conditions (like home or labs)

This clarity helps guide your design choices in the next steps.

IDEATE

In this phase, you start imagining how to solve the problem you just defined. *For our project, you came up with innovative idea like:*

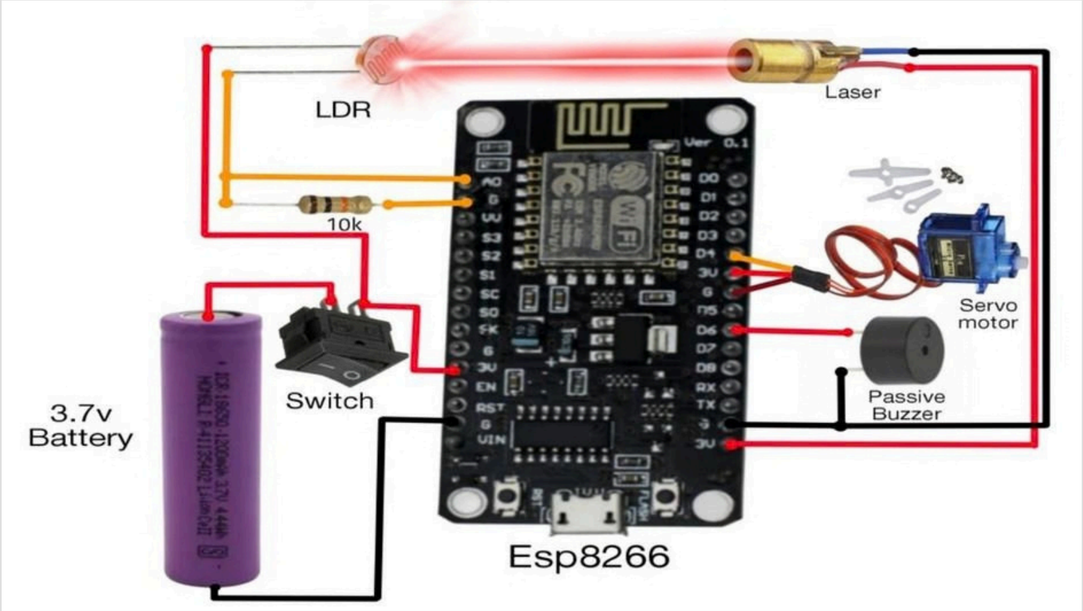
- Using a laser beam and LDR as an invisible tripwire to detect motion.
- Triggering a buzzer or mobile alert when the beam is broken.
- Making it user-friendly with simple controls or even smartphone access.
- Adding IoT features (like ESP8266) to send real-time notifications.
- Considering pet-safe modes and customizable zones for flexibility.

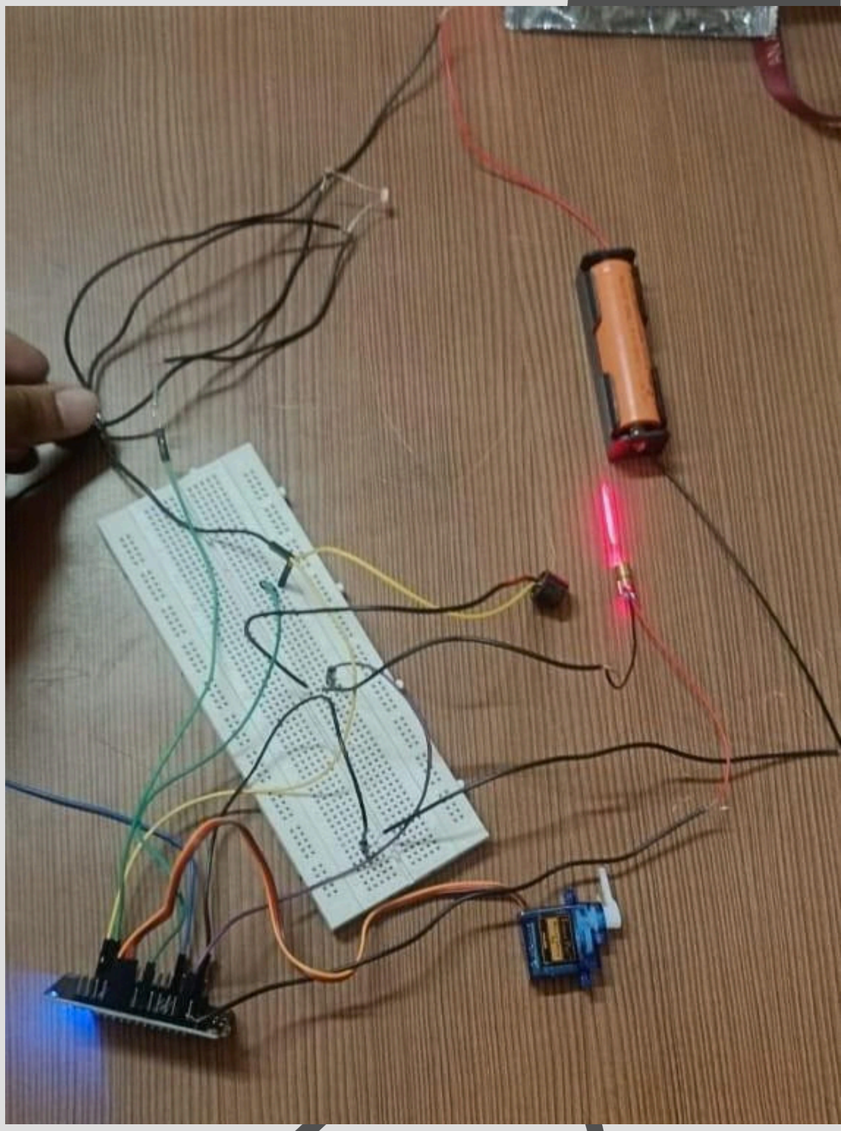
This step is about being creative—thinking beyond just the basic setup and finding ways to make the system more helpful, responsive, and inclusive.

PROTOTYPE

Prototype Build the system:

- Laser Diode: Acts as the emitter, projecting a continuous light beam.
- LDR (Light Dependent Resistor): Detects whether the laser beam is uninterrupted.
- Microcontroller (e.g., ESP8266 or Arduino): Processes the signal and controls the alarm.
- Buzzer & LED: Triggered when an intrusion is detected.



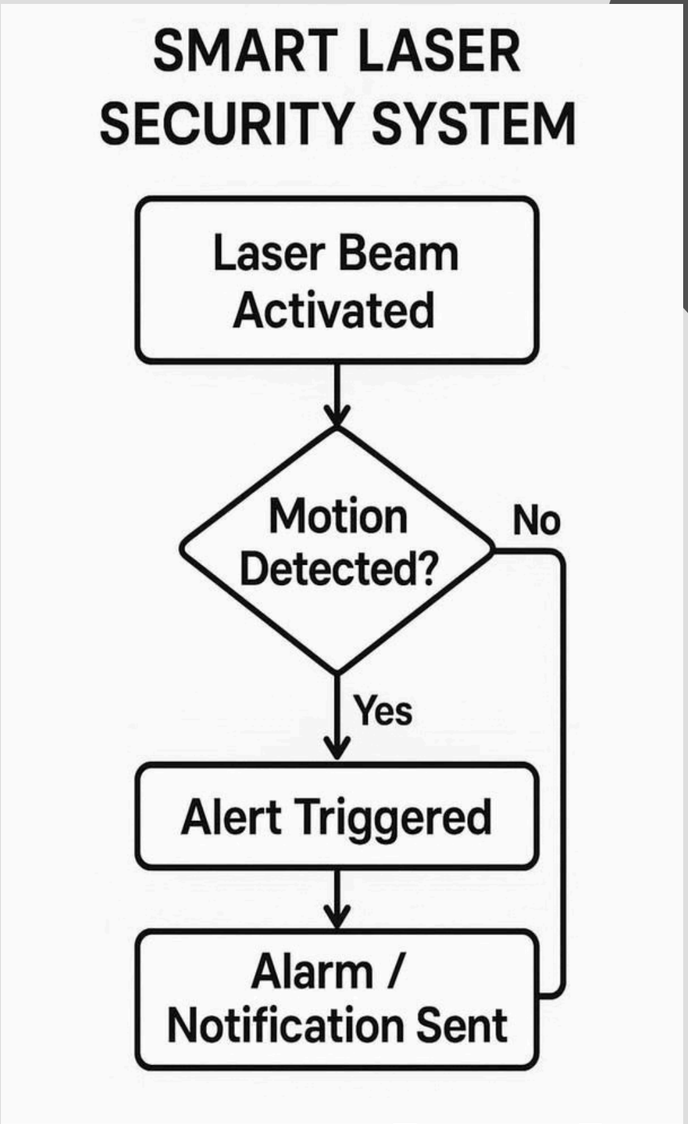


BLOCKDIAGRAM &FLOWCHART

BLOCK DIAGRAM

BLOCK DIAGRAM

- LaserDiodeemitsafocusedlightbeam.
- LDR/Photodiodedetectsthebeamcontinuously.
- Microcontroller monitors signal changes.
- Ifthe beam isbroken:
- Buzzer sounds an alert.
- LEDflashes for visual warning.
- Wi-FiModule sends notification to a phone (optional).



FLOWCHART

TESTING

- This step ensures your system works reliably under real-world conditions. Here’s how you test it: Normal Operation Test.”
  - Confirmed that the system stays idle when the laser beam is uninterrupted. No false alarms were triggered.
- Intrusion Detection Test
  - Blocked the laser beam using a hand or object.
  - Verified that the buzzer and/or LED activated instantly.
  - For IoT versions ,confirmed that mobile alerts were sent viaWi-Fi.
- Wi-FiAlertTest(OptionalIoTFeature)
  - Used platforms like Blynk or IFTTT. Checkedforreal-
  - timenotificationsonasmartphone. Measured response time
  - and reliability.
- 4.Repeatability & Stability Test
  - Repeated the intrusion test multiple times.
  - Ensuredconsistentperformanceandnomisalignmentofthelaser. Verified long-
  - termoperationwithoutsystemcrashesorfalsetriggers.

CONCLUSION

The laser-based detection system is an effective and reliable solution for security and automation applications. By utilizing an LDR to detect interruptions in a laser beam, coupled with a signal processing unit, it successfully triggers an alarm/buzzer to provide instant alerts. The addition of a servo motor introduces mechanical response capabilities, making it adaptable for applications like smart doors and robotic systems. Further enhancements, such as integrating an ESP8266, can enable wireless communication for real-time monitoring. Overall, this system demonstrates efficient detection, rapid response, and potential for IoT-based expansion, making it a versatile tool in modern technological applications.

THANK YOU